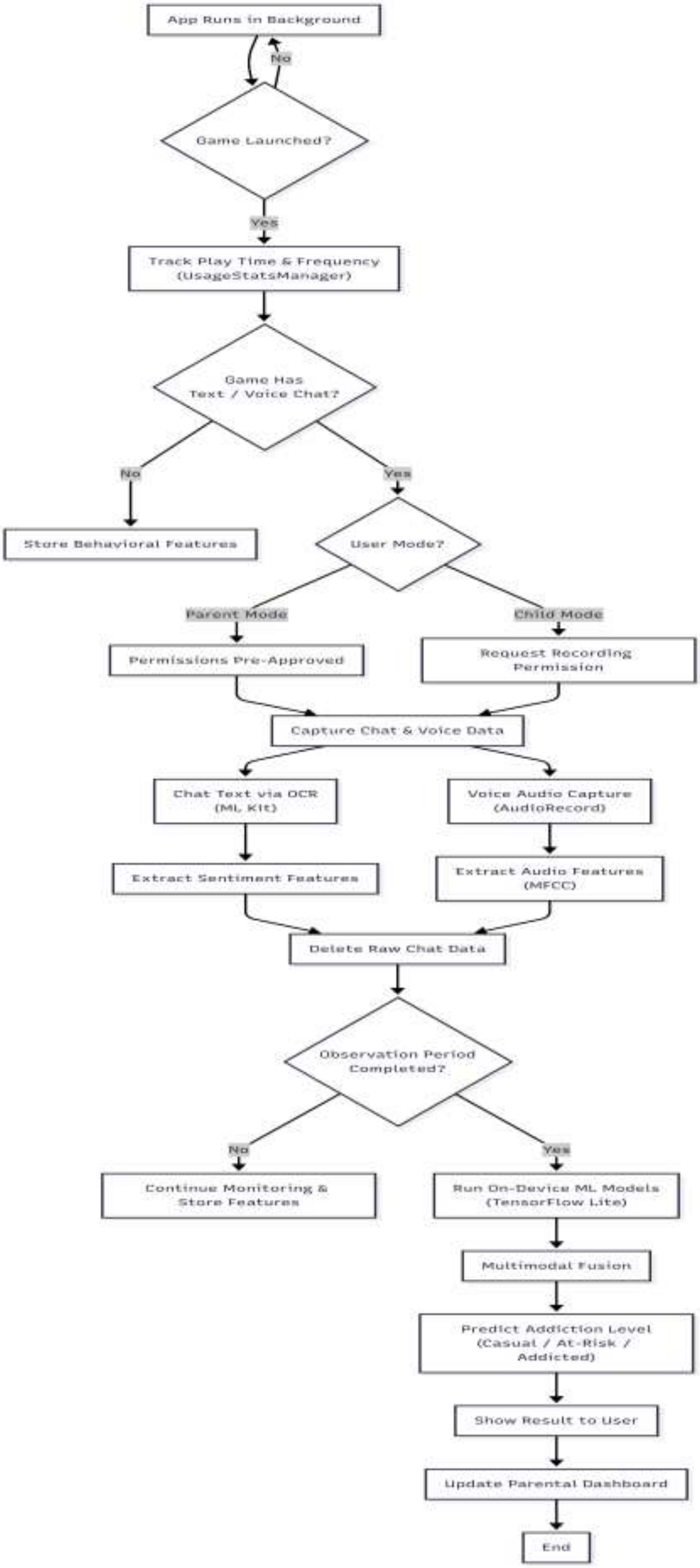
System Design and Android Implementation (APIs & Libraries)

The system is implemented as an Android application that runs in the background and detects when a game is launched using the **Android UsageStatsManager API**. This API is used to track playtime, frequency, and session patterns for all games, including casual ones like Sudoku.

For games that include text chat, the app captures the chat area using the **MediaProjection API** and extracts text with **Google ML Kit OCR**. Sentiment features are then analyzed using lightweight NLP models deployed on the device through **TensorFlow Lite**. For voice chat, audio is recorded using the **AudioRecord or MediaRecorder API**, and emotional features are extracted. During training, **Librosa** is used for MFCC extraction, and the trained speech emotion model is converted to **TensorFlow Lite** for on-device use.

All machine learning inference runs locally on the device using **TensorFlow Lite**, ensuring low latency and privacy. Raw chat and audio data are deleted immediately after processing. The system observes the user's gaming behavior for two to three weeks before making the first addiction prediction, which is shown to the user and summarized for parents through a separate dashboard.


App Runs in Background
No
Game Launched?
Yes
Track Play Time & Frequency (UsageStatsManager)
Game Has Text / Voice Chat?
No
Yes
Store Behavioral Features
User Mode?
Parent Mode
Child Mode
Permissions Pre-Approved
Request Recording Permission
Capture Chat & Voice Data
Chat Text via OCR (ML Kit)
Voice Audio Capture (AudioRecord)
Extract Sentiment Features
Extract Audio Features (MFCC)
Delete Raw Chat Data
Observation Period Completed?
No
Yes
Continue Monitoring & Store Features
Run On-Device ML Models (TensorFlow Lite)
Multimodal Fusion
Predict Addiction Level (Casual / At-Risk / Addicted)
Show Result to User
Update Parental Dashboard
End